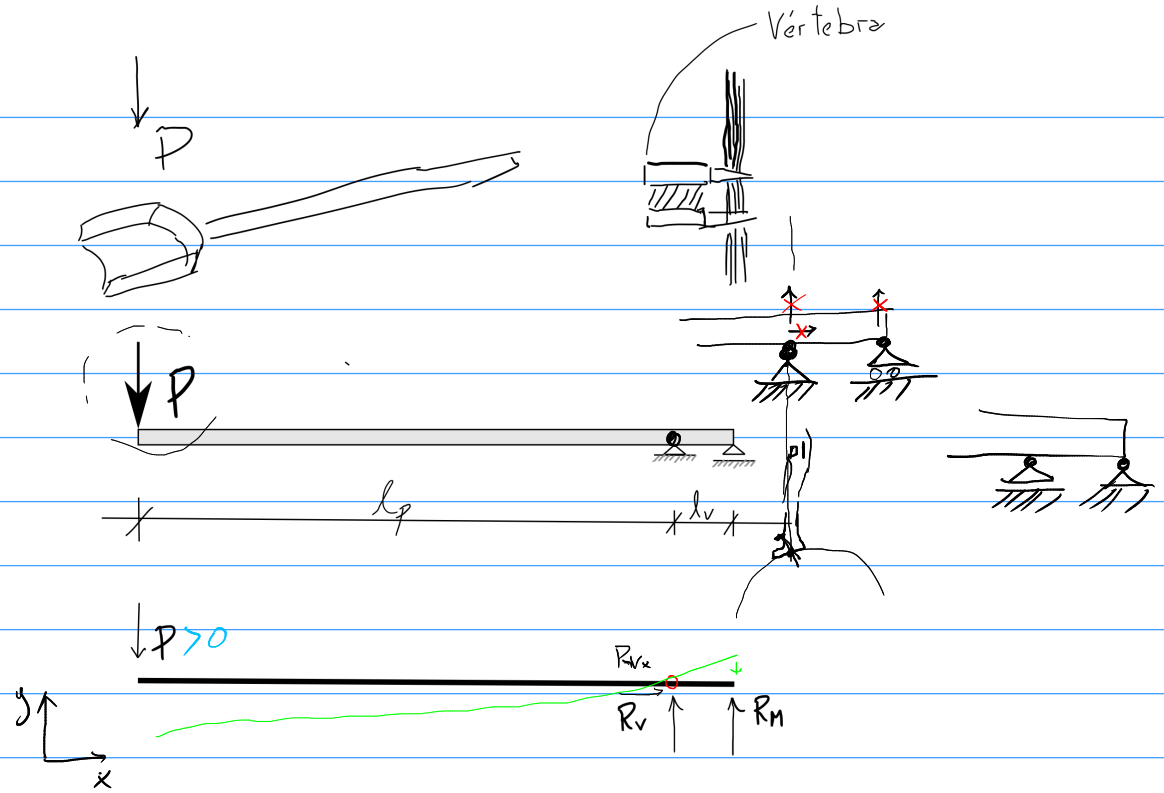


$l_p = 1m$ $\qquad$ $l_v \cong 0{,}02m$

$$P = m \cdot g = 10\,Kg \cdot 9{,}8\,\frac{m}{seg^2} = 98\,N$$

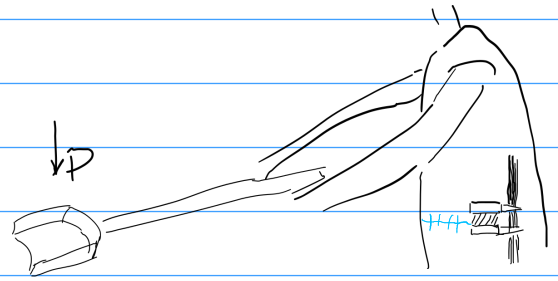
$\Sigma F_x = 0 \Rightarrow$ Verifica $\qquad R_{Vx} = 0$

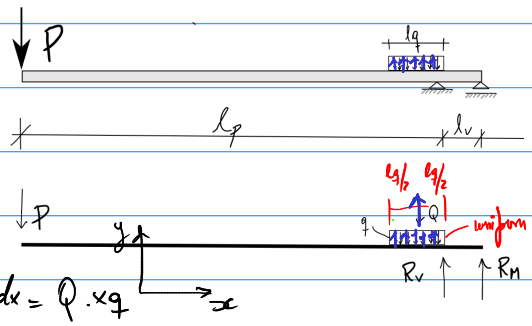
$\Sigma F_y = 0 \Rightarrow -P + R_V + R_M = 0$

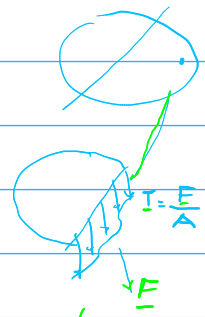
$\Sigma M = 0 \Rightarrow P l_p + R_M \cdot l_v = 0 \Rightarrow R_M = -\frac{P l_p}{l_v}$

$$R_V = P - R_M = P - \left(-\frac{P l_p}{l_v}\right) = P\left(1 + \frac{l_p}{l_v}\right)$$

$$R_V = 98N\left(1 + \frac{1m}{0{,}02m}\right) = 4998\,N$$







$$\underline{F} = \int_A \underline{\tau}\, dA$$

$$\underline{M} = \int_A \underline{r} \times \underline{\tau}\, dA$$

$$Q = \int_0^{l_q} \underbrace{q\, dx}_{dF}$$

$$M = \int_0^{l_q} dF . x = \int_0^{l_q} q . x\, dx = Q . x_q$$

$$\Sigma F_x = 0$$

$$\Sigma F_y = 0 = -P + Q + R_V + R_M$$

$$\Sigma M = P l_p - Q . \frac{l_q}{2} + R_M . l_v = 0 \Rightarrow R_M = \frac{1}{l_v}\left(Q \frac{l_q}{2} - P l_p\right)$$

$$R_V = \overbrace{P\left(1 + \frac{l_p}{l_v}\right)}^{>0} - \overbrace{Q}^{Q>0}\overbrace{\left(1 + \frac{1}{2}\frac{l_q}{l_v}\right)}^{>0}$$